Worksheet 5.4

# Precipitation

Student: ..... Class: .....

## Observing precipitation

**1.** The following photograph shows the result of adding a colourless solution of lead nitrate to a colourless solution of potassium iodide in a test tube.



(a) A yellow precipitate forms. Define the term ‘precipitate’.

(b) The beaker is allowed to stand for 1 hour. What will happen to the precipitate in that time?

(c) Name the compound making up the precipitate.

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(d) Write a word equation for the reaction.

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## Predicting precipitation

**2.** The following table provides information on the solubility of various salts in water.

| Anion | Soluble | Slightly soluble | Insoluble |
|---|---|---|---|
| Nitrate | All | None | None |
| Chloride | Most | $PbCl_2$ | AgCl |
| Sulfate | Most | $CaSO_4$, $Ag_2SO_4$ | $BaSO_4$, $PbSO_4$ |
| Carbonate | $Na_2CO_3$, $K_2CO_3$, $(NH_4)_2CO_3$ | None | Most |
| Hydroxide | NaOH, KOH, $NH_4OH$, $Ba(OH)_2$ | $Ca(OH)_2$ | Most |

Use this information to predict whether a precipitate will form when the nominated solutions are mixed. If a precipitate forms, name the precipitate.

(a) Lead nitrate solution + sodium sulfate solution.

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(b) Copper(II) sulfate solution + potassium hydroxide solution.

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(c) Barium chloride solution + calcium nitrate solution.

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